

Remote operation of connected, cooperative, and automated mobility services for goods transport.

Transportation and Logistics × Remote and autonomous operations × Edge computing · OS153 v1 · refreshed 2026-08-18

ATTRACTIVENESS	RIGHT TO WIN	COMPETITION	SERVICEABLE MARKET	HORIZON	PORTFOLIO
67 is the world moving	58 can we play, can we win	MEDIUM 0 named in the evidence	€824m partial evidence · per year	NOW budgeted procurement within 1...	L0 14 signals

Attractiveness and right to win are scored and displayed separately and are never combined (SC-12). Competition is a fourth quantity, not a discount on either.

The opportunity

This opportunity is about enabling remote operation of connected, cooperative, and automated mobility (CCAM) services for goods transport, leveraging edge computing to meet low-latency requirements. It targets transport operators and logistics companies that are beginning to integrate autonomous vehicles and need reliable, secure remote supervision and intervention. The timing is now because European research projects are advancing technical and societal readiness, and procurement signals show active investment in transport services.

Market opportunity

Method	Total (TAM)	Serviceable (SAM)	Obtainable (SOM)	Confidence
Bottom-up: enterprises x adoption x contract value	€1.3bn €290m – €7.7bn	€824m €180m – €4.8bn	€28.9m €6.3m – €168m	partial
Observed: tendered contract value, annualised	€115m	€115m	€4.0m	observed

All figures are annual, in EUR. Ranges compound the uncertainty of each factor below; the base case is the middle column of each.

Bottom-up: enterprises x adoption x contract value

Factor	Value	Basis	Source	As of
Enterprises in Transportation and Logistics (EU27_2020)	113,199 enterprises	observed	Eurostat · sbs_sc_ovw	2024
Enterprises adopting Edge computing	11.3 % of enterprises (10+ employees)	proxy	Eurostat · isoc_cicce_usen2 · E_CC_PCPU	2025
Annual value of one engagement	€792k per enterprise per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Size-weighted buyer base	14,748 enterprise-equivalents at large-enterprise engagement value	assumption	config/sizing.yaml · size_class_value_weights	size-2026-08-a
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Adoption rate is a declared proxy. Cloud computing services by NACE Rev. 2 activity series E_CC_PCPU is used as a proxy for Edge computing: Cloud compute purchase as nearest series. Contract values are observed from EU public procurement (TED). For private-sector verticals they are used as a proxy for comparable enterprise deal size: the buying entity differs, the scope of work is comparable. Engagement value is scaled by enterprise size class (10-19: x0.04, 20-49: x0.09, 50-249: x0.35, GE250: x1.0), because the observed contract value comes from large-organisation tenders.

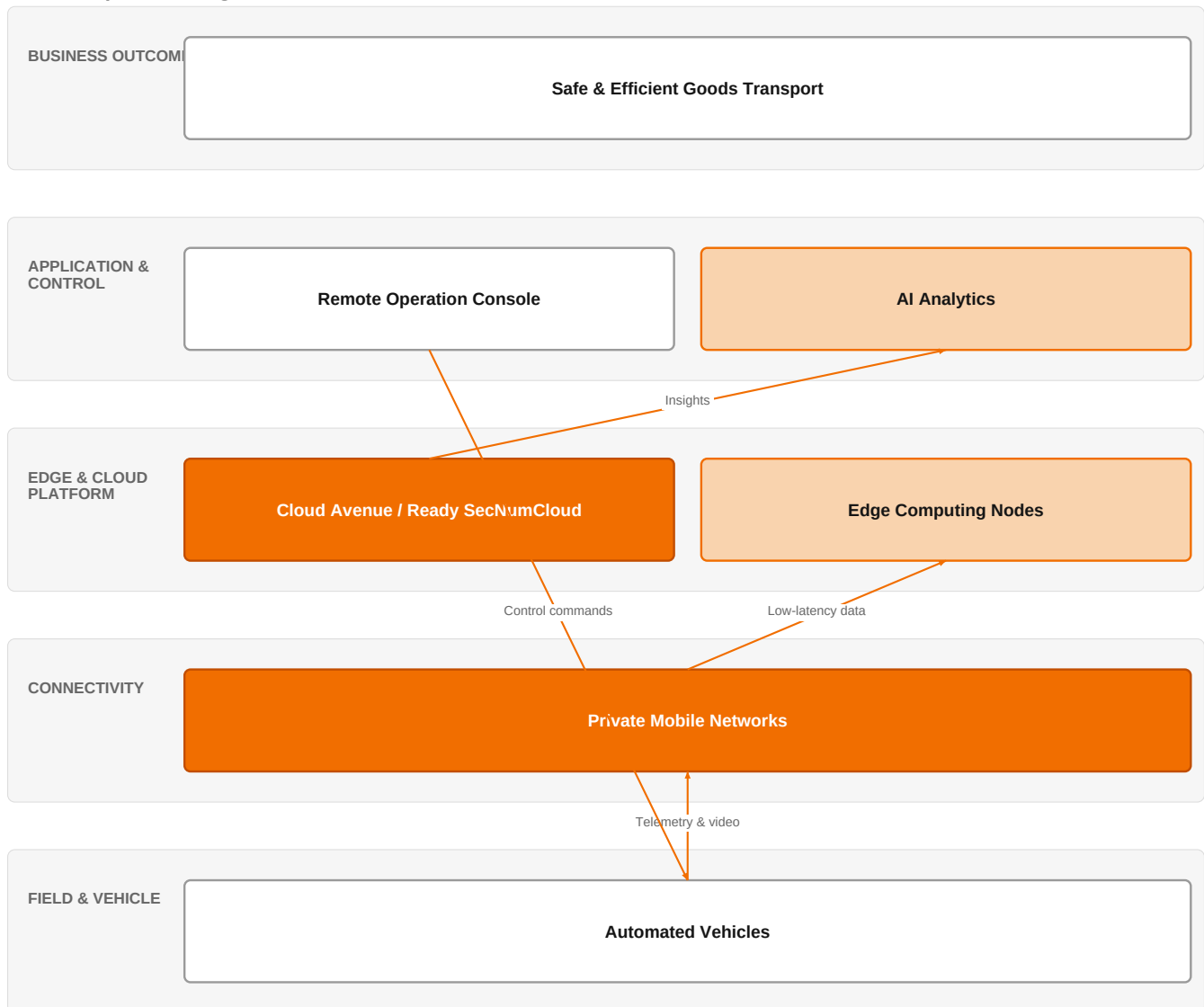
Observed: tendered contract value, annualised

Factor	Value	Basis	Source	As of
Tendered contract value in matching CPV categories	€115m per year	observed	TED (Tenders Electronic Daily) · ted	2026-05-26..2026-08-05
Assumed obtainable share of the serviceable market	3.5 % of SAM	assumption	config/sizing.yaml · obtainable_share	size-2026-08-a

Read this before quoting the number. Observed over 85 days of TED coverage and scaled to a year (x4.3); a short collection window makes that scaling rough. 8 matching notices, matched on vertical via the CPV crosswalk — §4.5.2's warning applies: a crosswalk error lands directly in this number. Public procurement only. Private-sector demand for the same capability is not visible here at all, so this is a floor rather than a market size.

The solution, in one picture

Remote Operation Edge Solution



■ Orange asset ■ Partner ■ Customer-owned ■ Third party / to be sourced

This diagram shows how Orange's private network and sovereign cloud enable secure, low-latency remote operation of automated vehicles.

Boxes marked as Orange assets are validated against the linked business graph; a box the model claimed for Orange without a matching asset is shown as third party.

What has changed

The integration of autonomous vehicles in transport systems has the potential to revolutionise goods transport as part of CCAM, but remote operation faces technical bottlenecks such as communication latency and the operator's inability to physically perceive environmental disturbances. Research is addressing these challenges with neural network-based teleoperation and cross-layer latency optimization for URLLC in 5G MEC networks. Additionally, societal readiness is being developed to ensure technological innovations are co-created with society, indicating a move toward acceptance and deployment.

Sources: SIG-10FAC6365ED9 cordis.europa.eu 2026-06-01; SIG-CDD2DA807ACB arxiv.org 2026-08-11; SIG-18012D9BCACF Springer Science and B 2026-06-19; SIG-D2F667783022 cordis.europa.eu 2026-05-01

Who buys, and why now

The primary buyer is the Chief Operating Officer or Head of Fleet Operations at a logistics or transport company, who feels the pain of driver shortages, rising costs, and the need for 24/7 operations. The trigger is often a pilot project or a regulatory push to adopt autonomous vehicles, or a competitive pressure to reduce operational costs. They need to ensure safety and efficiency while transitioning to automated fleets, and remote operation provides a fallback for handling edge cases and emergencies.

Why it is hot now — every claim bound to a dated source

- The integration of autonomous vehicles in transport systems has the potential to revolutionise the way we travel and transport goods as part of the Connected, Cooperative and Automated Mobility (CCAM) system. [SIG-10FAC6365ED9]
- The AcceptRemote project ensures that technological innovations in remote operations are co-created with and for society, developing a comprehensive Societal Readiness Framework. [SIG-D2F667783022]
- The integration of autonomous vehicles in transport systems has the potential to revolutionise the way we transport goods. [SIG-10FAC6365ED9]

What Orange would deliver, and why it can win

What Orange would deliver

Orange would assemble a solution combining its Private mobile networks for reliable, low-latency connectivity at depots and ports, and Cloud Avenue / Ready SecNumCloud for secure, sovereign edge computing. The engagement would start with a consultation using Orange Group researchers to assess the customer's specific remote operation needs, then deploy a private 5G network and edge cloud infrastructure, integrating partner technologies from NVIDIA, Dell, and HPE for AI and edge computing. The solution would be piloted with a specific use case, such as remote monitoring of automated trucks in a logistics hub.

Why Orange can win

Orange can win because it combines sovereign cloud (Cloud Avenue / Ready SecNumCloud) with private mobile networks, which is critical for trust and data security in transport operations. The partnership with NVIDIA, Dell, and HPE allows Orange to offer a comprehensive edge-to-cloud solution. However, the assets are not fully proven in this specific domain; Orange lacks a dedicated reference in remote CCAM operations, so the win must be built on the strength of its connectivity and cloud offerings.

Named assets behind this — each individually inspectable (LK-08)

Asset	Type	Link	Why it is linked	Curator
Private mobile networks	offer	L0	offer addresses the use case AND provides the technology	unconfirmed
Cloud Avenue / Ready SecNumCloud	offer	L1	offer provides the technology in a matching domain	unconfirmed
NVIDIA	partner	L2	partner provides the required technology	unconfirmed
Dell	partner	L2	partner provides the required technology	unconfirmed
HPE	partner	L2	partner provides the required technology	unconfirmed
AI / Data / Cloud experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
IoT experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Digital experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Defense and security experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Healthcare experts	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Orange Group researchers	capability pool	SUP	capability pool staffs this domain, vertical or technology	unconfirmed
Point One Navigation	reference	SUP	published reference in this vertical, different use case	unconfirmed

Competitive landscape

MEDIUM competitive intensity (score 7.1; 0 of 8 listed players appear in this topic's own evidence)

A real field, including at least one player the customer will already know. Expect to be compared.

Competitors include AWS, Deutsche Telekom / T-Systems, Lumen Technologies, NTT DATA, Schneider Electric, HPE, Nokia, and Vodafone Business. AWS and HPE are strong in edge computing, while Nokia and Vodafone are active in transportation and logistics. Deutsche Telekom and Lumen compete in connectivity solutions. The customer will compare on latency, reliability, security, and total cost of ownership. Orange's differentiator is its sovereign cloud and private

network integration, but it must prove its edge capabilities against these established players.

Sources: SIG-10FAC6365ED9 cordis.europa.eu 2026-06-01

Competitor	Category	Position here	Basis	Evidence
AWS	Hyperscaler	sells Edge computing; competes in Cloud — also an Orange partner	structural	—
Deutsche Telekom / T-Systems	Telecom operator peer	sells Edge computing; competes in Cloud, Connectivity Solutions	structural	—
Lumen Technologies	Telecom operator peer	sells Edge computing; competes in Connectivity Solutions	structural	—
NTT DATA	Global systems integrator	sells Edge computing; competes in Cloud, Connectivity Solutions	structural	—
Schneider Electric	Industrial / OT specialist	sells Edge computing	structural	—
HPE (Aruba, Athonet)	Network equipment vendor	sells Edge computing; competes in Connectivity Solutions — also an Orange partner	structural	—
Nokia	Network equipment vendor	sells Edge computing; active in Transportation and Logistics; competes in Connectivity Solutions	structural	—
Vodafone Business	Telecom operator peer	active in Transportation and Logistics; competes in Connectivity Solutions	structural	—

evidenced means this topic's own sources name the competitor — the signal ids are in the last column and are dated and clickable in the radar. **structural** means the curated register says they sell this technology into this vertical, which is true but is not proof they are in this deal. Register version competitors-2026-08-a.

Qualifying questions for the first meeting

- 1 Are you currently piloting or planning to pilot autonomous vehicles for goods transport in the next 12 months?
- 2 What is your biggest operational challenge with remote operation: latency, reliability, security, or something else?
- 3 Do you have existing private network or edge computing infrastructure at your key logistics hubs?
- 4 How do you currently handle fallback when autonomous vehicles encounter situations they cannot handle?
- 5 What are your data sovereignty requirements for operational data from your vehicles?

Objections you will meet

They will say	You can answer
We already have a cloud provider for edge computing.	That's true, but our private mobile network and sovereign cloud offer a level of security and low latency that general-purpose clouds may not provide for critical operations.
Remote operation is not mature enough for our use case.	You're right that it's evolving, but we can start with a pilot to prove the concept and build on the latest research and our partners' technologies.
We are concerned about the cost of deploying private 5G.	Cost is a valid concern, but the investment can be offset by operational efficiencies and the ability to scale. We can design a phased approach to manage costs.

Risks and unknowns

The main risk is that remote operation technology is still maturing; latency and reliability issues are not fully solved. There is also regulatory uncertainty around autonomous vehicle operations. The societal readiness is still being developed, which could slow adoption. Additionally, Orange's lack of a reference in this specific domain may make customers hesitant. The exact business case for remote operation in goods transport is not yet proven, and the cost of deploying private networks and edge infrastructure may be high.

Next action, by role (FR-17)

Role	Do this next
Presales / Proposal	Prepare a bid brief for a European logistics operator, positioning Orange's Private mobile networks and Cloud Avenue / Ready SecNumCloud alongside NVIDIA, Dell, and HPE partnerships to address edge computing requirements for remote goods transport, and incorporate the societal readiness angle from the AcceptRemote project.

Sales	In a conversation with a logistics company exploring autonomous trucking, mention Orange's Private mobile networks and Cloud Avenue / Ready SecNumCloud as foundational for secure, low-latency remote operations, and reference the collaboration with Point One Navigation as proof of our ecosystem's capability.
Strategist / Innovator	Commission a deep-dive brief on edge computing architectures for remote operation of goods transport, leveraging Orange's AI/Data/Cloud and IoT expertise, and assess the fit of Cloud Avenue / Ready SecNumCloud and Private mobile networks for this use case.

Evidence

Every claim in this brief resolves to one of these. Tier 1 is an authoritative primary source; tier 4 is vendor-published material, discounted in scoring (§4.3.7).

Signal	Date	Publisher	T	Title and link
SIG-E092212B9854	2026-08-05	ted.europa.eu	1	France – Public road transport services – Services de transports scolaires sur le ressort terri...
SIG-CF0145140CA8	2026-07-30	boamp.fr	1	Centrale telefonique de reservation MOOVICITE liee aux offres de transport sur reservation (TS...
SIG-18012D9BCACF	2026-06-19	Springer Science and Busi...	1	Cross-Layer Latency Optimization for URLLC Services in 5G MEC Networks Using Adaptive Network S...
SIG-1B99E37EFA33	2026-06-10	ted.europa.eu	1	Norway – Transport services (excl. Waste transport) – Transport service for adapted school tran...
SIG-B0AAA55D7DB2	2026-06-09	boamp.fr	1	Diffusion des editions d'information et de promotion du reseau regional des transports de Prove...
SIG-10FAC6365ED9	2026-06-01	cordis.europa.eu	1	Advancing Technical and Societal Readiness of Remote Operation in Connected, Cooperative, and A...
SIG-8D368CE4ADA7	2026-05-27	ted.europa.eu	1	Spain – Non-scheduled passenger transport – SERVICIO DE TRANSPORTE INTERCENTROS ENTRE LOS HOSPI...
SIG-D2F667783022	2026-05-01	cordis.europa.eu	1	Acceptance of Remote Operations for Future Mobility
SIG-BA196FF05244	2026-02-03	Springer Science and Busi...	1	Digital Product Passport for Circular Economy: A Review of Semantic Web Vocabularies and Requir...
SIG-81CEE24B054D	2025-12-12	openalex.org	1	Blockchain-Driven Supply Chain Automation for Real-Time Traceability
SIG-40DC35D6534F	2025-10-14	openalex.org	1	Predictive Generation of Digital Product Passport as Decision Support in Circular Networks
SIG-1373FFA5C063	2025-07-01	cordis.europa.eu	1	Holistically Optimised Logistics Operations and Green Innovative Solutions for Transport Cargo ...
SIG-C0910B893CD2	2024-06-01	cordis.europa.eu	1	Preparing the ground for AUTonomous Multimodal SUPply Chains
SIG-CDD2DA807ACB	2026-08-11	arxiv.org	3	A Neural Network Based Teleoperation for Remote Controlled Vehicles

How this brief was made

Computed, not written. Attractiveness, right to win, competitive intensity and every figure in the market-opportunity section are computed from stored inputs and can be re-derived (DR-05, NFR-01). No language model produced a number anywhere in this document (§4.4.4).

Written, then checked. The narrative sections and the solution diagram were generated from this topic's evidence and from the named asset and competitor lists. Factual sections must cite signal ids attached to this topic; sections that failed that check, or that contained a generated quantity or an unsupplied organisation, were removed rather than rewritten.

What	Version	When
Scores — weight set	w-2026-08-a	2026-08-18T19:08:26+00:00
Market sizing — assumptions	size-2026-08-a	2026-08-18T21:06:14+00:00
Competitor register	competitors-2026-08-a	2026-08-18T21:06:18+00:00
Narrative — prompt / model	describe-v1 / deepseek-chat	2026-08-19T04:57:27+00:00
Pipeline	0.1.0	2026-08-18

Generated 2026-08-19 04:58 UTC. Scores computed under a different weight set are not comparable with these (SC-10), and neither are market sizes computed under a different sizing version.